

24CS31 — 3-HOUR HAIL MARY · v2, corrected

Organised **unit by unit**, with a map of exactly how each question will be printed and which parts you can answer. **Steps 1-7 are unchanged from v1** — nothing you have already done is wasted.

WHAT CHANGED AND WHY. Re-running the optimiser with *nothing* pre-cut found three topics I had wrongly dropped. All three beat items I had kept:

- **ODE / simultaneous ODE by Laplace (07)** — 4/4 papers and **+7 on every single one**. The only topic that gains on all four. Cutting it was my biggest mistake.
- **SVD (10)** — 4/4, **no calculus at all** (it is $A^T A$, eigenvalues, eigenvectors), and the **same matrix twice** including the most recent paper. It passes your own rule and I cut it anyway.
- **Rotation-operator proof (07)** — **pure bookwork, zero calculus, 12 minutes**, and it completes Q5 in Unit III. I dismissed it on a 2/4 frequency without noticing it is free marks.

Result: the worst-case floor rises from 59 to 65 (72 with overtime) in the same three hours.

THE CLOCK — study in this order

stop anywhere; the total is the worst of 4 papers

#	Study this	Unit	Appears as	Marks	Mins	Clock	Running total
1	Unitary / Hermitian check	V	Q10b	05	08	0:08	5
2	Prove T is linear + find images	III	Q5b	07	12	0:20	12
3	Least-squares solution	IV	Q8b	10	20	0:40	22
4	t^n -multiplication proof	I	Q1c / Q2b	07	15	0:55	29
5	Heaviside piecewise + LT	II	Q3c	07	20	1:15	36
6	$L\{\sin \sqrt{t}\} \rightarrow L\{\cos \sqrt{t}/\sqrt{t}\}$	I	Q1b	04	12	1:27	40
7	The Unit I 2-mark definition	I	Q1a / Q2a	02	05	1:32	42
8	ODE / simultaneous ODE by LT ● ADDED	II	Q3d / Q4d	07	22	1:54	49
9	Singular Value Decomposition ● ADDED	V	Q9c	10	22	2:16	54 ✓ PASS
10	Orthogonal diagonalization	V	Q10c	10	25	2:41	59
11	Span / basis / linear independence	III	Q5a	06	12	2:53	64
12	Rotation-operator proof ● ADDED	III	Q5c	07	12	3:05	65
— 3 HOURS UP — overtime below takes the floor to 72 —							
13	Positive-definite check	V	Q10a	05	08	—	completes Unit V's 5+5 → Q10 becomes a full 20
14	Composition of transformations	III	Q6a	06	12	—	backs up Unit III if the rotation proof is absent
15	Kernel & range + rank-nullity	III	Q6c	07	15	—	with the above, Q6 becomes an alternative 13
16	$L^{-1}\{\log(\dots)\}$	II	Q3b	04	12	—	+4 in Unit II
17	The Unit II 2-mark definition	II	Q3a / Q4a	02	05	—	+2, five minutes — Q3 becomes a full 20
18	Four fundamental subspaces	IV	Q7c	07	15	—	+10 in Unit IV — always in Q7
19	Complete solution of $Ax = b$	IV	Q7a	06	15	—	with the above, Q7 becomes a full 20
20	Transition matrix	III	Q6b	07	15	—	completes Q6 as a second full 20

STILL CUT — and now for the right reasons. Each is either long with values that change every paper, or adds **nothing** to your worst case because it sits in a question you would not answer:

QR factorization (4/4, but the matrix changed in all four papers — +0 to the floor) • **convolution** (sits opposite Heaviside) • **periodic functions** (+0 — it is always in the question you are *not* answering) • improper integrals $\int_0^\infty$ • $L\{t e^{at} \text{ trig}\}$ / division by t • $(\sqrt{t}+1/\sqrt{t})^3$ • diagonalize + A^n • Gram-Schmidt • orthogonal projection • Markov • quadratic forms • PCA.

IN THE HALL. Read both questions in a unit, tick the parts you can do, answer the one with more ticks. **Write every part you know and never leave a unit blank.** Answer the parts, not the question.

UNIT I — Laplace Transforms

20 marks · Q1 or Q2 · a(02)+b(04)+c(07)+d(07) · you score **13**

	a — definition	b — short result	c — proof	d — long evaluation
Q1	(02) Write the LT of a periodic function step 7 ✓	(04) Given $L\{\sin \sqrt{t}\}$, show the $\cos \sqrt{t}$ result step 6 ✓	(07) t^n-multiplication proof step 4 ✓	(07) $\int_0^\infty$ integral + $L\{\sin 3t \cos 2t\}$ — cut ✗
Q2	(02) Define the Laplace transform step 7 ✓	(04) $L\{(\sqrt{t}+1/\sqrt{t})^3\}$ — cut ✗	(07) $L\{te^{-4t}\sin 3t\}$ / division by t — cut ✗	(07) Periodic function — cut ✗

→ **ANSWER Q1 — 13 of 20**. Trigger: the t^n proof. In all four papers it sits in the **same** question as the $\sin \sqrt{t}$ result, and the periodic-function problem sits in the **other** one.

4 Prove the t^n -multiplication rule

07 marks · Q1c or Q2b — **4/4 papers, word for word**

If $L\{f(t)\} = F(s)$, prove $L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} \{F(s)\}$, n a positive integer.

Appears alongside: the $\sin \sqrt{t}$ result (step 6) — **always**. 11 marks in one question.

Four lines. $F(s) = \int_0^\infty e^{-st} f(t) dt$. Differentiate under the integral: $F'(s) = -L\{tf(t)\}$. Each further differentiation drops another $(-t)$. Close with **induction on n** , base case $n=1$.

6 Given $L\{\sin \sqrt{t}\}$, show the $\cos \sqrt{t}$ result

04 marks · Q1b — same statement in Jun 23 and Mar 24

Given $L\{\sin \sqrt{t}\} = \frac{\sqrt{\pi}}{2s^{3/2}} e^{-1/4s}$, show $L\{\frac{\cos \sqrt{t}}{\sqrt{t}}\} = \frac{\sqrt{\pi}}{s^{1/2}} e^{-1/4s}$.

Appears alongside: the t^n proof (step 4).

$\frac{d}{dt}(\sin \sqrt{t}) = \frac{\cos \sqrt{t}}{2\sqrt{t}}$, so the target = **2 × $L\{\text{that derivative}\}$** . Apply $L\{f\} = sF(s) - f(0)$, $f(0) = 0$. **Three lines.**

7 The Unit I 2-mark definition

02 marks · Q1a and Q2a — **one in each**

“Write the LT of a periodic function” • “Define the Laplace transform”

Appears alongside: whichever question you choose.

Periodic, period T : $\frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt$. **Definition:** $\int_0^\infty e^{-st} f(t) dt$. **Formula, not a paragraph.**

UNIT II — Application of Laplace Transforms

20 marks · Q3 or Q4 · you score **14**, or **20** with overtime

	a — definition	b — short inverse	c — main problem	d — ODE / application
Q3	(02) Dirac-delta definition overtime 17 ✗	(04) $L^{-1}\{\log(\dots)\}$ overtime 16 ✗	(07) Heaviside piecewise + LT step 5 ✓	(07) LR-circuit current step 8 ✓
Q4	(02) Unit-step definition overtime 17 ✗	(04) Partial-fraction inverse — cut ✗	(07) Verify the convolution theorem — cut ✗	(07) Simultaneous ODEs by LT step 8 ✓

→ **ANSWER Q3 — 14 of 20, and a full 20 after two 17-minute overtime items**. This is the unit the correction fixed: the **d** part — an ODE, a system of ODEs, or the circuit problem — appears in **both** questions in every paper, so step 8 pays whichever you pick.

5 Heaviside piecewise + Laplace transform

07 marks · Q3c — same function in Apr 23 and Mar 24

Express $f(t) = \begin{cases} t^2, & 0 < t < 2 \\ 4t, & 2 < t < 4 \\ 8, & t > 4 \end{cases}$ via the Heaviside function; hence find its Laplace transform.

Appears alongside: the ODE / circuit problem in the **d** slot (step 8) — in all four papers. **That pairing is 14 marks.**

1 — stack the jumps: $f = t^2 + (4t - t^2)u(t-2) + (8-4t)u(t-4)$. **2 — rewrite each bracket in $(t-a)$:** $4 - \tau^2$ and $-8 - 4\tau$. **3 — second shifting:** $L\{f\} = 2/s^3 + e^{-2s}(4/s - 2/s^3) + e^{-4s}(-8/s - 4/s^2)$

8 Solve ODEs and simultaneous ODEs by Laplace transforms

07 marks · Q3d or Q4d — 4/4 papers · the item I wrongly cut

(a) **Simultaneous** — $\frac{dx}{dt} - 2y = \cos 2t$; $\frac{dy}{dt} + 2x = \sin 2t$,

$x(0)=1$, $y(0)=0$.

(b) **Single ODE** — $y'' + 4y' + 3y = e^{-t}$ (set twice, with different initial conditions).

(c) **Circuit** — $L \frac{di}{dt} + Ri = Ee^{-at}$, $i(0)=0 \rightarrow i = \frac{E}{R-aL}(e^{-at} - e^{-Rt/L})$.

Appears alongside: the Heaviside question (step 5) in the same question — 4/4 papers.

One method for all three. Transform every term with $L\{y'\} = sY - y(0)$ and $L\{y''\} = s^2Y - sy(0) - y'(0)$. For a single ODE, solve for $Y(s)$ and invert by partial fractions. For a system, you get two linear equations in $X(s)$ and $Y(s)$ — solve them as simultaneous algebra, then invert each. **The initial conditions do the work; write them in at the transform step and the rest is algebra.**

UNIT III — Vector Space and Linear Transformation

20 marks · Q5 or Q6 · you score 13-20

	a — vector space (06)	b — transformation (07)	c — the long one (07)
Q5	(06) Span / basis / independence step 11 ✓	(07) Prove T is linear + images step 2 ✓	(07) Rotation-operator proof step 12 ✓
Q6	(06) Composition of transformations overtime 14 ✗	(07) Transition matrix overtime 20 ✗	(07) Kernel & range overtime 15 ✗

→ **ANSWER Q5 — a full 20** once step 12 is done. That is what adding the rotation proof buys: Q5's three parts become $6 + 7 + 7 = 20$. The proof appeared in Apr 23 and Mar 24; in the other two papers Q5c was the kernel/transition question instead, which is why **overtime 14-15 is your insurance**.

2 Prove T is linear + find images

07 marks · Q5b — all four papers, same slot

(a) $T(x, y) = (3x + y, 2y, x - y)$; images of $(1, 2)$ and $(2, -5)$.

(b) $T: P_2 \rightarrow P_2$, $T(ax^2 + bx + c) = (a + b)x^2 + c$; image of $5x^2 + 6x + 1$.

Appears alongside: span/basis at Q5a and the rotation proof at Q5c — the three together are Q5's full 20.

Show $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ and $T(k\mathbf{u}) = kT(\mathbf{u})$, then substitute. **Answers:** $T(1, 2) = (5, 4, -1)$, $T(2, -5) = (1, -10, 7)$, $T(5x^2 + 6x + 1) = 11x^2 + 1$.

11 Span / basis / linear independence

06 marks · Q5a — all four papers

Is $\left\{ \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 3 & 4 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ 1 & 2 \end{bmatrix} \right\}$ a basis of M_{22} ? or do $\{(1, 2, 3), (-1, -1, 0), (2, 5, 4)\}$ span R^3 ?

Appears alongside: the prove- T -is-linear question at Q5b.

One method for every version: flatten each object to a coordinate vector (a 2×2 matrix $\rightarrow$ a 4-vector), stack as rows of a square matrix, take the **determinant**. Non-zero = independent = basis / spans.

12 Rotation-operator proof

07 marks · Q5c — Apr 23 and Mar 24 · **the item I wrongly cut**

Show that the linear operator $T: R^2 \rightarrow R^2$ defined by $T(\mathbf{x}) = A\mathbf{x}$ rotates $\mathbf{x}$ through an angle θ about the origin, where $A =$

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Appears alongside: the span/basis and prove-linear questions — **this is the part that turns Q5 into a full 20**.

Pure bookwork — twelve minutes, no calculus, 7 marks. Write $\mathbf{x}$ in polar form: $\mathbf{x} = (r\cos\varphi, r\sin\varphi)$. Multiply out $A\mathbf{x}$ and you get $(r\cos\theta\cos\varphi - r\sin\theta\sin\varphi, r\sin\theta\cos\varphi + r\cos\theta\sin\varphi)$. Apply the compound-angle formulae $\rightarrow (r\cos(\theta + \varphi), r\sin(\theta + \varphi))$. Same modulus r , argument increased by θ — so it is a rotation through θ . State linearity ($T(\mathbf{x}) = A\mathbf{x}$ is linear for any matrix A) and you are done.

UNIT IV — Orthogonal Projections

20 marks · Q7 (06+07+07) or Q8 (10+10) · you score 10, or 20 with overtime

	a	b	c
Q7	(06) Complete solution of $Ax = b$ overtime 19 (06) ✗	(07) Orthogonal projection — cut (07) ✗	(07) Four fundamental subspaces overtime 18 (07) ✗
Q8	(10) QR factorization — cut (10) ✗	(10) Least-squares solution step 3 (10) ✓	(—) — ✗

→ **ANSWER Q8 — 10 of 20 from one part.** Overtime 18 and 19 turn **Q7 into a full 20** and are the best-paying overtime on the sheet (+10 in this unit). QR stays cut: 4/4, but its matrix changed in every single paper, so it adds **nothing** to your worst case.

3 Least-squares solution of $Ax = b$

10 marks · Q8b — all four papers

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix} \quad b = \begin{bmatrix} 3 \\ 5 \\ 7 \\ -3 \end{bmatrix} \quad \text{— identical in Jun 23 and Mar 24}$$

Appears alongside: the QR factorization at Q8a, which you are skipping.

Solve the normal equations $A^T A \mathbf{x} = A^T b$. That is the whole question. $A^T A = \begin{bmatrix} 4 & 8 & 10 \\ 8 & 20 & 26 \\ 10 & 26 & 38 \end{bmatrix}$, $A^T b = \begin{bmatrix} 12 \\ 12 \\ 20 \end{bmatrix} \Rightarrow \hat{\mathbf{x}} = (10, -6, 2)$.

Substitute back to check.

UNIT V — Applications of Eigenvalue Decomposition

20 marks · Q9 or Q10 · a(05)+b(05)+c(10) · you score 15-20

	a — 5-mark check	b — 5-mark check	c — the 10-marker
Q9	(05) Quadratic form — <i>cut</i> ✗	(05) Markov steady state — <i>cut</i> ✗	(10) SVD step 9 ✓
Q10	(05) Positive-definite check overtime 13 ✗	(05) Unitary / Hermitian check step 1 ✓	(10) Orthogonal diagonalization step 10 ✓

→ **ANSWER Q10 — 15 of 20, a full 20 after overtime 13 (eight minutes).** Adding SVD covers the *other* question too: in Apr 23 orthogonal diagonalization and SVD were **both in Q10**, which is exactly why cutting SVD cost 10 marks on that paper — the worst one.

1 Unitary / Hermitian check

05 marks · Q10b — a property check appears in **all four papers**

Is $A = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}$ unitary? or is $\begin{bmatrix} 3 & 7-4i & -2+5i \\ 7+4i & -2 & 3+i \\ -2-5i & 3-i & 4 \end{bmatrix}$ Hermitian?

Appears alongside: the positive-definite check — the property checks travel as a pair in all four papers.

Write $A^* =$ conjugate of the transpose. **Hermitian** $\Leftrightarrow A^* = A$; **unitary** $\Leftrightarrow A^*A = I$. **Both answer YES.** Show A^* in full.

9 Singular Value Decomposition

10 marks · Q9c — **4/4 papers · same matrix twice · the item I wrongly cut**

Find the SVD of $A = \begin{bmatrix} 1 & 1 \\ 3 & -3 \end{bmatrix}$ — identical in Jun 23 and Mar 24

Appears alongside: the quadratic-form and Markov 5-markers (Mar 24 Q9) — or, in Apr 23, orthogonal diagonalization itself.

No calculus at all — it is eigenvectors twice over. $A^T A = \begin{bmatrix} 10 & -8 \\ -8 & 10 \end{bmatrix}$, eigenvalues **18 and 2**, so the singular values are $\sigma_1 = 3\sqrt{2}$, $\sigma_2 = \sqrt{2}$. Eigenvectors of $A^T A$ are $(1, -1)/\sqrt{2}$ and $(1, 1)/\sqrt{2} \rightarrow$ those are the columns of V . Then $u_i = Av_i / \sigma_i$ gives the columns of U , and $\Sigma = \text{diag}(3\sqrt{2}, \sqrt{2})$. Write **$A = U\Sigma V^T$** .

10 Orthogonal diagonalization

10 marks · **Q10c — same matrix in 3 of the 4 papers**

Orthogonally diagonalize $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$

Appears alongside: the two 5-mark property checks — in 3 of 4 papers they share this question.

You already know the answer — rehearse producing it. Eigenvalues **2, 3, 6**, all distinct, so the eigenvectors are automatically orthogonal — **no Gram-Schmidt**, and saying so earns marks. Eigenvectors $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$; normalise by $\sqrt{2}, \sqrt{3}, \sqrt{6}$ to get the columns of P . Then **$P^T A P = \text{diag}(2, 3, 6)$** . Show the three dot products are 0.

13 Positive-definite check

05 marks · Q10a — same matrix in Apr 23 and Mar 24

Is $A = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}$ positive definite?

Appears alongside: the unitary/Hermitian check (step 1).

Leading principal minors are **1, 0, 0 = NOT positive definite**; it is positive *semi*-definite (row 3 = row 1, eigenvalues 0, 0, 6). **Say that explicitly.** Eight minutes, and it completes Q10 as a full 20.

Why this version differs from v1: the first pass pre-cut every calculus-heavy topic before optimising. Re-running with nothing pre-cut showed three of those cuts were wrong — the ODE question gains 7 marks on *every* paper, SVD is calculus-free and repeats verbatim, and the rotation proof is 7 marks of bookwork that completes Q5. Worst-case floor: **59 → 65**, and **72** with overtime. Steps 1-7 are unchanged. Working in *coverage_audit.py*.